Mohammed Mahdi Jahangiri

AI Software Developer / Forward-Deployed Engineer

Cardiff, UK | mapjiv@live.com | +44 7473 252475
Portfolio | linkedin.com/in/m-mahdi-jahangiri | github.com/devPirate01 | huggingface.co/devPirate01

AI Software Engineer with 2+ years of cross-functional production delivery. Highly fluent in spec-first, AI-assisted development to accelerate shipping without sacrificing architectural rigor. Specialise in LLM integration, evaluation pipelines, and building robust REST APIs to bridge AI models with existing systems. Proven ability to walk into ambiguous problems and engineer pragmatic, scalable solutions, recently raising team sprint success from 40% to 90%. Passionate about translating complex business logic into maintainable, secure-by-design software.

Technical Skills

Core Languages: Python, C#, C++
ML & AI: LLM integration, prompt engineering, evaluation pipelines, RAG, PyTorch, Hugging Face, scikit-learn
Engineering: FastAPI, REST APIs, JSON data plumbing, Git, CI/CD, Agile/Scrum, Object-Oriented Architecture

Selected Projects & Publications

Deterministic LLM Orchestrator & Legacy Systems

Python, C++, QLoRA, JSON APIs

- Architected an asynchronous **Python** middleware bridging a stateful legacy **C++** engine with local edge LLMs, designing strict JSON schema validators and fallback state machines to eliminate hallucinated system states
- Fine-tuned a **Gemma 2B** model via **QLoRA** 4-bit quantisation on consumer hardware, lifting rule-routing accuracy from **19.0% to 53.4%** ($p < 0.001$) while outperforming a 26B baseline at **sub-second latency (0.98s)**
- Developed an automated empirical evaluation harness across **1,500 multi-seed trials**, validating strict epistemic boundaries (< 1% leakage), shipping weights to **Hugging Face**, and deploying a live mod to **Nexus Mods**

Real-Time AI Interaction Middleware

Python, FastAPI, C++, Hugging Face

- Built a local **Python FastAPI** server acting as the data plumbing layer, securely bridging real-time input to an external client application via a custom REST API
- Integrated **Whisper**, **Wit.ai** and local **Hugging Face** models for robust text/voice intent detection

Cross-Org Systems Engineering — Awkward Studios

C#, Applied Mathematics, API logic

- Walked into a messy, ambiguous problem with **no existing spec or tooling**, engineering a custom mathematical correction system for venue partner **CULTVR** and sponsor **Media Cymru**

End-to-End Production App — Meta Quest Store

C#, Meta Quest SDK, Delivery Pipelines

- Owned end-to-end delivery of a production software application: build deployment, strict platform compliance review, store approval, and iteration based on real user performance metrics

Publication — ICHORA 2024

LLM and Pursuit Learning Automata for Dialogues

- Published and presented a pragmatic, resource-efficient local LLM pipeline, applying **pursuit learning automata** to balance user satisfaction against processing constraints

Experience

Engineering Team Coordinator — Landell Games

Feb 2025 – Present

Remote / Sweden — Promoted from Engineering Intern (Dec 2024 – Jan 2025). Team lead on a live-service software project.

- Owned weekly build pipelines and enforced software engineering best practices (Git branch hygiene, code reviews) across **5 department leads**
- Coordinated a **3–6 person engineering team**, translating messy business priorities into shipped features and raising sprint success from **40% to 90%**
- Made pragmatic architectural decisions, profiling system bottlenecks and improving frame time by **10–30 FPS**

Education

MSc Artificial Intelligence — Cardiff University

2025–2026

Dissertation completed: safe AI companions using fine-tuned local LLM inference.

- Computer Vision & NLP: bounding-box localisation via **RF-DETR** on **VOC2012**; NLP pipelines with **pandas/scikit-learn** for data gathering, labelling, and intent/emotion analysis
- **AI Safety Founding Fellow (Spring 2026):** Completed fundamentals fellowship; evaluated catastrophic AI risks, compliance, model evaluation frameworks, and secure-by-design AI governance

BEng Computer Engineering — Azad University, Tehran

2020–2025

- Graduated with **First-Class Honours**; core study: **Data Structures & Algorithms**, **Advanced Programming (C++)**, **SQL** and database design, Computational Architecture and Networks

Awards & Interests

DiffTrailblazers — Cardiff University: Won 1st place, receiving a **£1,500** prize for a practical innovation project.

Interests: Judo (green belt) | Mandarin Chinese | Playing with AI agents and automation